Natural Language Processing

A subfield of Artificial Intelligence (AI) that deals with the processing of Natural Language.

The term Natural language refers to languages in common and daily use.

The study of Natural Language Processing (NLP) is concerned with developing intelligent systems that can be augmented with capabilities for processing Natural Language.

NLP enables computers to understand, interpret, generate, and respond to human language in both text and speech.

NLP combines computer science, linguistics, and machine learning to process language. A typical NLP system performs tasks such as:

Text preprocessing: Cleans and prepares text (removing punctuation, tokenization, stemming).

Language understanding: Identifies grammar, meaning, context, and relationships between words.

Machine learning or deep learning: Learns patterns from large amounts of language data.

Response generation: Produces meaningful outputs, such as translations, summaries, or answers.

Common Applications:

Translating (Arabic to English),

Chatbots and virtual assistants,

Sentiment analysis, Spam email detection, Speech recognition, Text summarization, Question answering systems, Autocomplete and predictive text, etc.

Specifically, it is the process of a computer extracting meaningful information from natural language input and producing a natural language output. In theory, natural language processing is a very attractive method of human-computer interaction.

Natural language understanding (NLU) is referred to as an AI-complete problem because it requires extensive knowledge of common-sense reasoning, context understanding, ambiguity resolution, reasoning and inference, knowledge of the world, pragmatics and intention. Language understanding is deeply connected to human intelligence. To understand language as people do, an AI must not only process words and grammar but also reason, use common sense, interpret context, and apply real-world knowledge. This makes NLU one of the most challenging problems in artificial intelligence.

Natural Language Processing

Processing written text: using lexical, syntactic, and semantic knowledge of the language as well as the required real world information.

Processing spoken language: using all the information needed above plus additional knowledge about phonology as well as enough added information to handle the further ambiguities that arise in speech.

Subfields of NLP

Natural Language Understanding

Concerned with the understanding of knowledge specified in Natural Language such that the computer can use it to perform it tasks.

It is much easier to build a computer system that can do calculus than to build one that can understand language. The reason is that understanding language requires a vast amount of knowledge.

Natural Language Generation

Challenges in Natural Language Understanding

The Problem:

people often leave out information because they assume the listener already knows the context. As a result, sentences are incomplete descriptions of the full meaning they are intended to convey **(incompleteness)**

There are some people outside

- some people are in the lawn? Outside the door? On the street?

John arrived late

- Late for what? (a meeting, class, work, family gathering)

I'll be there in a minute.

- Where is "there"? Does "a minute" literally mean one minute or simply "soon"?

I called Nancy to ask her to go shopping. She said she'd love to go.

- She answered the phone? Are you both going? She will come here to go?

The Good Side:

Language allows speakers to be as vague or as precise as they want; leaving things on the listener to understand or at times known already.

The Problem:

Based on the underlying context, the same expression can mean different things **(lexical ambiguity)**

The Good Side:

Language lets us communicate about an infinite world using a finite (learnable) number of symbols.

The Problem:

New words, phrases, abbreviations, slang, and technical terms are regularly introduced, while the meanings of existing words can also change over time. This makes it difficult for NLP systems to maintain an up-to-date understanding of language.

- Technology terms: Cloud weather phenomenon but today also means internet-based storage and computing services
- Social media vocabulary: Selfie, Hashtag, Unfollow
- Internet slangs: LOL, DM, Ghosting
- New AI-related words: Prompt engineering, Hallucination, Generative AI

Impact on NLU systems:

- Unknown words: The system may not recognize newly created words.
- Incorrect interpretation: It may assign an outdated meaning to a word.
- Reduced accuracy: translation, sentiment analysis, question answering become less reliable.
- Frequent updates required: Language models and dictionaries need continual retraining or updating to keep pace with language changes

The problem:

The same meaning or information can be expressed using many different words, phrases, or sentence structures.

While humans easily recognize different expressions conveying the same idea, computers struggle unless trained to recognize paraphrases and synonyms.

Different wording, same meaning

John purchased a car.
John bought a car.
John got himself a new car.

Active vs. Passive voice

The teacher praised the student.
The student was praised by the teacher.

Challenges in Natural Language Understanding

Ambiguity:

- Lexical Ambiguity: Multiple meaning of the same word

The pitcher is empty
The pitcher is angry
She filled the pitcher with cold lemonade.
The pitcher trapped insects inside its leaves.
The startup's pitcher impressed the investors with a clear presentation

That's a good book.
This soup is so good.
Her initiative received a good response.

She is a bright student.
She closed her eyes against the bright sunlight.
She greeted us with a bright smile.
Fury turned her face bright red.
She always looks on the bright side

Syntactic ambiguity: I hit the man with the hammer

How do you interpret this sentence?
Did I hit the man who was holding the hammer or did I hit the man using a hammer?

The chicken is ready to eat.
Is the chicken going to eat, or is it cooked and ready to be eaten?

Referential Ambiguity: Unclear antecedents, problem in understanding target of pronouns such as:

John hates Bill because ***he*** sympathized with Sam
who is he? Does **he** imply John or Bill?
John told David that he had won the prize.
Who won the prize: John or David?

Imprecision: consider the following sentences

I waited for the taxi for a long time.
The meeting lasted a long time.
He died because he didn't eat for a long time.

Here the term long time has imprecise meaning, in one sentence it may mean several minutes whereas in the other one it probably means several days.

Inaccuracy: refers to situations where a sentence contains incorrect, false, or misleading information. The language itself may be grammatically correct, but the facts it expresses are wrong or inconsistent with reality.

An NLU system must distinguish between understanding the sentence and determining whether its content is accurate.

Inaccuracy can rise due to following factors

- Spelling errors
- Incomplete sentences
- Improper punctuation, etc.

Solution:

Some possible solutions to the discussed stated problems:

Use of background knowledge and context
Relate idea to a familiar situation
Infer based on recognition of familiar pattern

ANALYSIS LEVELS OF NATURAL LANGUAGE UNDERSTANDING

Natural Language Understanding involves several complex processes. To manage this complexity, linguists have defined different levels of analysis for natural language:

- Phonology
- Morphology
- Syntax
- Semantics
- Pragmatics
- Discourse Integration
- Prosody

Phonology

Phonology is the branch of linguistics that studies the sound system of a language: how speech sounds (phonemes) are organized and used to distinguish meaning.
In NLU, phonology is important because spoken language must first be interpreted correctly before its meaning can be understood.

Word Boundary Detection ("I scream" vs. "ice cream")
Handling Pronunciation Variations ("tomato" may be pronounced differently in American and British English)
Homophone Recognition (two, too, to OR flower, flour)

Morphology

It looks at the ways in which word break down into components and how that affects their grammatical status.

Syntax

It involves applying the rules of the grammar from the language being used.
It determines the role of each word in a sentence, and enable computers to convert into easily manipulated structures.
It requires some kind of parsing method.

Semantics

It involves the examination of the meaning of words and sentences.

Some methods of semantic analysis makes use of various types of grammars, which are formal systems of rules that attempt to describe the ways that the sentences can be constructed.

Semantic Analysis must do two important things:

- It must map individual words into appropriate objects in the knowledge base or database.
- It must create the correct structures to correspond to the way the meanings of the individual words.

Such systems include Conceptual dependency theory and Conceptual Networks.

Pragmatics

Pragmatics is the branch of linguistics that studies how context influences the meaning of language. It helps a system understand the speaker's intended meaning, which may be different from the literal meaning of the words.

For example: “Can you tell me the time?”

- Here “Yes” is not the suitable answer

It's cold in here

- Depending on the context, it may mean: Please close the window. Please turn on the heater. Or maybe the speaker is simply making an observation.

I am starving.

- I’m very hungry and want to eat now.

Great! Nice job!!

- Usually Praise but after a mistake may express sarcasm, depending on context.

Discourse Integration

- It deals with the rhythm and intonation of language. Or, The stress and intonation patterns of an utterance.
- This level of analysis is difficult to formalize and often neglected.

"I never said she stole my money"

Depending on which word is emphasized, this sentence could have several distinct meanings:

"**I** never said she stole my money" - Someone else said it, but **I** did not.

"I **never** said she stole my money" - I simply didn't ever say it.

"I never **said** she stole my money" - I might have implied it in some way, but I never explicitly stated it.

"I never said **she** stole my money" - I said someone took it; I didn't say it was her.

"I never said she stole my money" - I just said she probably borrowed it.

"I never said she stole my money" - I said she stole someone else's money.

"I never said she stole my money" - I said she stole something, but not my money.

Natural Language Generation

It is the area of NLP concern with making it easier for us to understand a computer output.

The ability of computer to generate the output in our natural language allows more understandability of the computer's response.

A natural language generation program has the following three basic functionalities:

- The program must decide when to say something
- The program must decide what to say.
- The program must decide how to say it.

Natural Language Processing System

Parsing

Parsing ensures that the computer understands the precise function of each word in a sentence and its relation with other words in the sentence.
In this process sentence is converted into a hierarchical structure that corresponds to the units of meaning in the sentence.

Parser

The parser is a piece of software that analyzes the input sentence syntactically.
Each word is isolated and its part of speech is identified.
The parser then maps the words into a structure called the parse tree.
The parse tree shows the meanings of all of the words and how they are assembled.

The job of the parser is to examine each word in a sentence and create the parse tree that identifies all the words and puts them together in the right way.

Lexicon

The parser and the lexicon work together to pick apart a sentence and then create the parse tree. In operation, the parser is largely a pattern matcher. Once the individual word is identified, the parser searches through the lexicon comparing each input word with all of the words stored there. If a match is found, the word is put aside along with the other lexical information., such as part of speech and meaning.